

# **Design Document**

## **HALO: High Autonomous Low-SWaP Operations**

### **Team Members**

**Sloan Hatter** ([shatter2022@my.fit.edu](mailto:shatter2022@my.fit.edu))

**Blake Gisclair** ([bgisclair2022@my.fit.edu](mailto:bgisclair2022@my.fit.edu))

### **Faculty Advisor**

**Dr. Ryan T. White** ([rwhite@fit.edu](mailto:rwhite@fit.edu))

**September 29, 2025**

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# 1. Introduction

## 1.1. Purpose

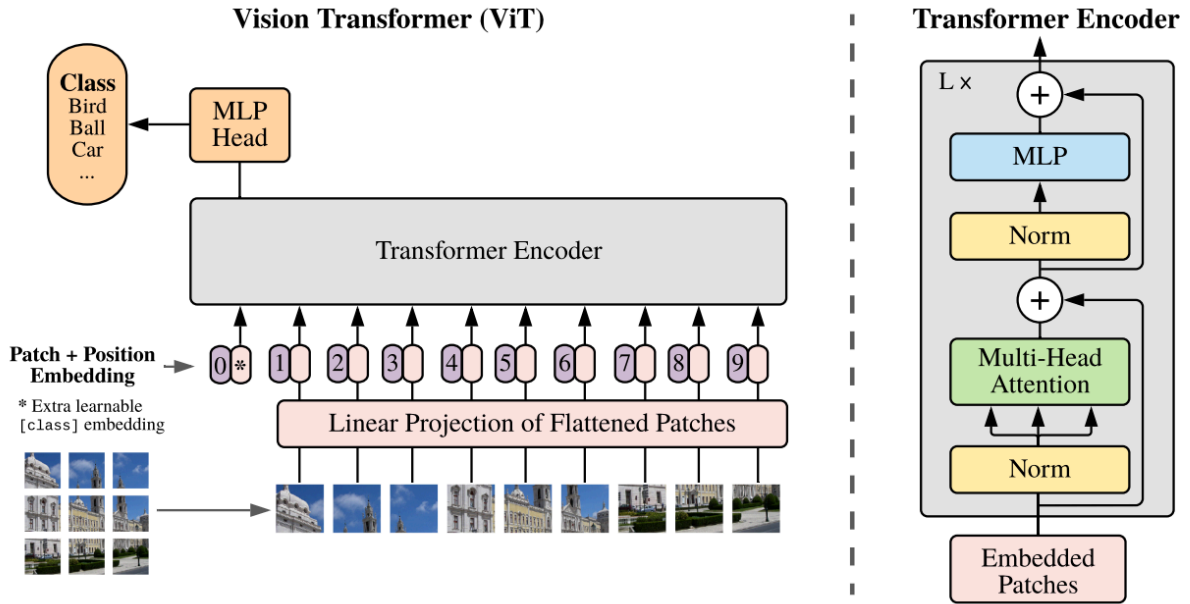
The purpose of this Design Document is to describe the architecture, components, and design decisions for the HALO system through system architecture diagrams and schematics.

## 1.2. Scope

This document covers the design and architecture of HALO, including its neural network model, data processing pipeline, and deployment platform.

# 2. System Design

## 2.1. Vision Transformer Model



\* weights restricted to the set  $\{0,1\}$

Image to Patches

- The  $H \times W \times C$  size image is split into non-overlapping  $P \times P$  patches. Each patch is flattened into a vector of length  $P^2 C$ . This converts the 2-D image into a token sequence of length  $N = \frac{HW}{P^2}$  that the transformer can read.

Patch Embedding

- Each flattened patch in  $\mathbb{R}^{P^2 C}$  is linearly projected to a  $D$ -dimensional token  $x_i = W_{EPI}$ , which produces an  $N \times D$  matrix. This places all patches into a common feature space of width  $D$  for later comparisons.
- 1-Bit: Use a binary  $W_E \in \{0,1\}^{D \times P^2 C}$  and compute with bitwise AND +

popcount, then multiply by a learned scale per output channel.

#### CLS Token

- A learned vector  $x_{cls} \in \mathbb{R}^D$  is prepended, which makes the sequence length  $N + 1$ . This token aggregates global information that the classifier reads at the end.

#### Position Embeddings

- A position code  $p_i$  is added to every token in order to restore patch layout as Transformers do not know order by default.

#### Embedded Token Sequence

- The standardized sequence  $X \in \mathbb{R}^{(N+1) \times D} : [x_{cls}, x_1, \dots, x_N] + pos$  is derived; this is the input to all encoder blocks.

#### Transformer Encoder Block

- Each block applies Normalization  $\rightarrow$  Multi-Head Self-Attention  $\rightarrow$  Residual Add  $\rightarrow$  Normalization  $\rightarrow$  MLP  $\rightarrow$  Residual Add to Whole Sequence.
- Normalization

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